Lecture 25

Topological band theory

Objectives. *Lecture 24 attached a line bundle to an isolated simple eigenvalue. For a periodic crystal, the parameters are momenta on a torus. The normalized curvature integral on that torus is an integer. For a two-band model we compute it as a degree, then use the cited Hall-response theorem to relate this integer to a measured conductance.*

By the end of this lecture you will be able to:

- Construct a Bloch eigenline bundle for a smooth isolated simple band.
- Prove integrality of its normalized curvature flux on the oriented momentum torus.
- Compute the lower-band Chern number and gap closings of the QWZ model.
- State the filled-band Hall formula with its physical hypotheses and component convention.

25.1 Bloch Hamiltonians and the momentum torus

Consider a square lattice with cell label $r \in \mathbb{Z}^2$ and N orbitals in each cell. Use lattice spacing one and a fixed orthonormal orbital basis. A one-particle wavefunction has a column $\psi_r \in \mathbb{C}^N$ at each cell. A translation-invariant finite-range Hamiltonian is given by

$$(H\psi)_r = \sum_{R\in\mathbb{Z}^2} T_R \psi_{r+R}, \qquad T_{-R} = T_R^\dagger,$$

where only finitely many hopping matrices T_R are nonzero. The adjoint condition makes the operator Hermitian. Recall the positive-phase Bloch convention from *Quantum Mechanics I*, Lecture 8. Substitution of $\psi_r = e^{\mathrm{i}\mathbf{k}\cdot r} v$ gives the *Bloch Hamiltonian*

$$H(\mathbf{k}) = \sum_R T_R e^{\mathrm{i}\mathbf{k}\cdot R}.$$

It is a smooth Hermitian $N \times N$ matrix. Adding 2π to either momentum component leaves this matrix unchanged. Thus its parameter space is the *Brillouin torus* $\mathbb{T}^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2$, oriented by $\mathrm{d}k_x \wedge \mathrm{d}k_y$.

On an $L \times L$ periodic lattice, the allowed momenta have components $2\pi a/L$, with $a \in \{0, \dots, L-1\}$. The finite geometric sum makes Bloch waves with distinct momenta orthogonal. There are L^2 such momenta, each paired with the N orbital columns. The resulting Bloch waves form a basis of the NL^2-dimensional one-particle space. The substitution above therefore gives its full block decomposition. The continuous torus describes the same matrix family as the system size grows. The infinite-lattice Bloch ansatz is not a normalized square-summable state.

A *band* is an energy branch $E_n(\mathbf{k})$ of this family. We call a chosen smooth simple band *isolated* when, at every momentum, its distance from every other eigenvalue is at least a

fixed $\delta > 0$. This is a separation at the same momentum. It does not yet assert an energy interval separating occupied and empty states at all momenta.

Proposition 25.1 (The Bloch bundle · *proved*)**.** *The eigenspaces of a smooth isolated simple band form a Hermitian line bundle $L_n \to \mathbb{T}^2$. For local unit eigenvectors $v_i(\mathbf{k})$, its connection is $\mathcal{A}_i = v_i^\dagger \mathrm{d}v_i$. Its* band Chern number *is*

$$C_n = \frac{\mathrm{i}}{2\pi} \int_{\mathbb{T}^2} \mathrm{d}\mathcal{A}.$$

Proof. The periodic matrix family is defined on the torus itself. Lecture 24's adjugate construction supplies smooth local eigenvectors there. Their unit-phase transitions give the line bundle and the projected Hermitian connection. Its curvature is global and imaginary, so the displayed integral is real. The next proposition proves that it is an integer. ∎

The square in the figure is a parameter domain. Its edge frames need not agree.

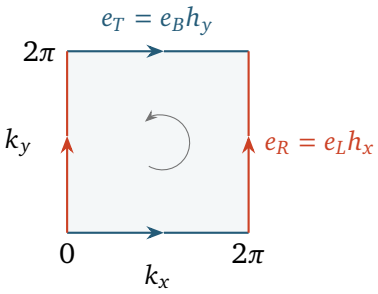


Figure 25.1: Opposite sides represent the same momenta. The circular arrow gives the positive torus orientation. The two unit-phase functions compare the chosen square frame on each pair of edges.

25.2 Integral flux and the two-band degree

A *smooth family* of Hermitian line bundles with connection here means one Hermitian line bundle with a smooth unitary connection on $\mathbb{T}^2 \times [0,1]$, restricted to the momentum slices.

Proposition 25.2 (Integrality on the torus · *proved*)**.** *For a Hermitian line bundle with a unitary connection on the oriented torus, the number $\mathrm{i}\int_{\mathbb{T}^2} F/(2\pi)$ is an integer. It is unchanged in a smooth family as defined above.*

Proof. Pull the bundle and connection back along $\mathbb{R}^2 \to \mathbb{T}^2$. Transport a fixed unit vector at the origin along radial segments. Lecture 21's smooth parameter dependence gives a smooth unit frame on $\mathbb{R}^2$. Restrict it to the square $[0, 2\pi]^2$ and call its connection form A. Write $x = k_x$ and $y = k_y$ in this argument. After identifying equal torus fibres, the edge frames satisfy

$$e(2\pi, y) = e(0, y) h_x(y), \qquad e(x, 2\pi) = e(x, 0) h_y(x),$$

with smooth maps $h_x, h_y \colon [0, 2\pi] \to \mathrm{U}(1)$. Lecture 3's phase lifting on an interval gives real smooth functions χ_x and χ_y with $h_x = e^{\mathrm{i}\chi_x}$ and $h_y = e^{\mathrm{i}\chi_y}$. Smoothness follows locally by a smooth inverse of the exponential covering. The frame law gives $A_R - A_L = \mathrm{i}\mathrm{d}\chi_x$ along

the vertical edges and $A_T - A_B = \mathrm{i}\mathrm{d}\chi_y$ along the horizontal edges. With $\Delta\chi = \chi(2\pi) - \chi(0)$, Stokes therefore gives

$$\int_{\mathbb{T}^2} F = \mathrm{i}(\Delta\chi_x - \Delta\chi_y), \qquad \frac{\mathrm{i}}{2\pi}\int_{\mathbb{T}^2} F = \frac{\Delta\chi_y - \Delta\chi_x}{2\pi}.$$

The right and top comparisons give the same frame at the last corner, so

$$h_y(0)h_x(2\pi) = h_x(0)h_y(2\pi).$$

Substitution of their lifted phases makes $\Delta\chi_x - \Delta\chi_y$ an integer multiple of 2π. This proves integrality. Neither edge phase is required to be a closed loop. In such a family, its global curvature restricts smoothly to the momentum slices, so the curvature integral varies continuously. An integer-valued continuous function on an interval is constant, proving the last claim. ∎

For a smooth Hamiltonian family whose chosen band stays isolated, the adjugate construction is smooth jointly in momentum and deformation parameter. The eigenline and projected connection of the chosen band give such a family.

For an example of integrality, the square potential $A = -\mathrm{i}n x\mathrm{d}y/(2\pi)$, with $n \in \mathbb{Z}$, has $h_x(y) = e^{-\mathrm{i}ny}$ and $h_y(x) = 1$. Its flux is $-2\pi\mathrm{i}n$, so its Chern number is n. The edge phase at $y = 2\pi$ agrees with its value at zero precisely because n is an integer.

Now take two orbitals per cell and a Hamiltonian $H(\mathbf{k}) = \mathbf{d}(\mathbf{k})\cdot\boldsymbol{\sigma}$, where $\mathbf{d}\colon \mathbb{T}^2 \to \mathbb{R}^3 \setminus \{0\}$ is smooth. The Pauli product identity gives energies $-|\mathbf{d}|$ and $+|\mathbf{d}|$. Define the direction map $\widehat{\mathbf{d}} = \mathbf{d}/|\mathbf{d}|$.

Theorem 25.3 (The lower band as a degree · *proved*). *For the lower band and the outward orientation on S^2,*

$$\mathrm{d}\mathcal{A} = -\frac{\mathrm{i}}{2}\widehat{\mathbf{d}}\cdot(\partial_{k_x}\widehat{\mathbf{d}}\times\partial_{k_y}\widehat{\mathbf{d}})\mathrm{d}k_x\wedge\mathrm{d}k_y, \qquad C = \deg(\widehat{\mathbf{d}}).$$

Proof. The lower eigenspace of $\widehat{\mathbf{d}}\cdot\boldsymbol{\sigma}$ is the excited eigenspace of Lecture 24's $-\mathbf{n}\cdot\boldsymbol{\sigma}$. That sphere line has curvature $-\mathrm{i}\beta/2$, where β is the outward area form. The band line is its pullback by $\widehat{\mathbf{d}}$. Pullback of the projected connection pulls back its curvature, giving the displayed formula. Lecture 14's degree identity gives

$$C = \frac{1}{4\pi}\int_{\mathbb{T}^2}\widehat{\mathbf{d}}^{\,*}\beta = \deg(\widehat{\mathbf{d}}),$$

because $\int_{S^2}\beta = 4\pi$. ∎

The upper band has the opposite curvature and Chern number. For more than two orbitals, the curvature definition still applies, but this particular map to a direction sphere is unavailable.

25.3 The Qi–Wu–Zhang model

The *Qi–Wu–Zhang model*, abbreviated QWZ, has direction vector

$$\mathbf{d} = (\sin k_x, \sin k_y, u + \cos k_x + \cos k_y), \qquad u \in \mathbb{R}.$$

Its Hamiltonian is the matrix

$$H(\mathbf{k}) = \begin{pmatrix} u + \cos k_x + \cos k_y & \sin k_x - \mathrm{i} \sin k_y \\ \sin k_x + \mathrm{i} \sin k_y & -u - \cos k_x - \cos k_y \end{pmatrix}.$$

The parameter u is an energy parameter, distinct from Lecture 19's inverse radius. In the lattice convention of the first section its only possibly nonzero hopping matrices are $T_0 = u\sigma_z$, $T_{(1,0)} = (\sigma_z - \mathrm{i}\sigma_x)/2$ and $T_{(0,1)} = (\sigma_z - \mathrm{i}\sigma_y)/2$, together with their adjoints. Thus it is a finite-range Hermitian lattice system.

The figure marks the momenta at which both transverse components vanish.

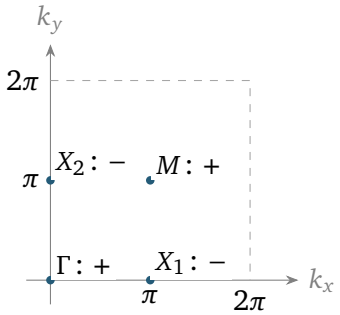


Figure 25.2: The four distinct vertical points in one torus cell. The signs are the local orientation signs used when the corresponding mass is positive.

The next figure records the lower-band integer. The missing endpoints are gap closings.

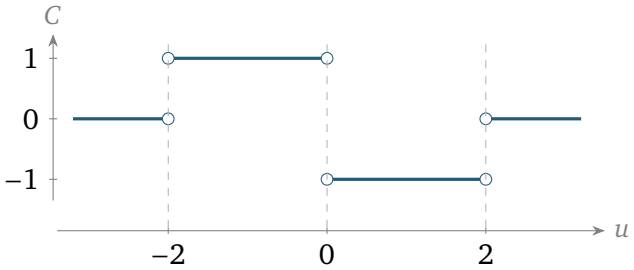


Figure 25.3: The QWZ lower-band Chern number as u increases. The number is undefined at $u = -2$, $u = 0$ and $u = 2$.

Example 25.4 (The QWZ phase diagram · *proved*)**.** Find the lower-band Chern number and every gap closing as u varies. The energies are $E_- = -|\mathbf{d}|$ and $E_+ = |\mathbf{d}|$, so a gap closes exactly when $\sin k_x = \sin k_y = 0$ and $u + \cos k_x + \cos k_y = 0$. The relevant masses and transverse Jacobian signs are

point	(k_x, k_y)	d_z	$\cos k_x \cos k_y$
Γ	$(0, 0)$	$u + 2$	$+1$
X_1	$(\pi, 0)$	u	-1
X_2	$(0, \pi)$	u	-1
M	(π, π)	$u - 2$	$+1$

Hence the critical parameters are -2, 0 and 2. There are two distinct gap-closing points at zero. At every other parameter, compactness of the torus gives a positive minimum gap.

To compute the degree, count preimages of the north pole. A row contributes precisely when its mass d_z is positive. Near such a point the first two components of $\widehat{\mathbf{d}}$ are local sphere

coordinates. Their Jacobian determinant is $\cos k_x \cos k_y / d_z^2$ at the point. These coordinates have the outward sphere orientation at the north pole. All selected preimages are therefore regular, with the signs in the last column. Lecture 14's signed-count theorem gives

$$C = \begin{cases} 0, & u < -2, \\ +1, & -2 < u < 0, \\ -1, & 0 < u < 2, \\ 0, & u > 2. \end{cases}$$

For $u < -2$ there are no north preimages. In the next interval only Γ contributes. Crossing zero adds both negatively oriented points. Crossing two adds the positive contribution at M. Thus the simultaneous closing at zero changes the integer by -2.

These values use $C = (\mathrm{i}/2\pi) \int \mathrm{d}\mathcal{A}$. Asbóth–Oroszlány–Pályi's continuum convention uses $Q = -C$.[1] This sign difference is not a reversal of the Hamiltonian or the torus orientation.

The exact degree calculation requires no numerical mesh. An independent numerical check may use forward overlaps of neighbouring lower-band eigenvectors. With a counterclockwise plaquette, the product of normalized overlaps has argument approaching $\mathrm{Im} \int \mathrm{d}\mathcal{A}$. Thus the *negative* sum of these arguments, divided by 2π, approximates our C. At $u = -3, -1, 1, 3$, grids with 20×20 and 32×32 points give values close to $0, +1, -1, 0$, respectively, in agreement with a separate curvature integral. These numerical checks do not replace the degree proof.

25.4 Filled bands and the Hall response

At zero temperature, noninteracting electrons fill all states below the *Fermi energy* E_F. An insulator has an open energy interval around E_F containing no bulk states. This full energy gap is stronger than separation of a band at each momentum. A *Chern insulator* is a band insulator whose occupied bands have nonzero total Chern number.

Let $e > 0$ be the magnitude of the electron charge, so its signed charge is $-e$. Write Planck's constant as $h_{\mathrm{P}} = 2\pi\hbar$, distinct from a frame-change map. Fix the scalar Hall conductance by the transverse component convention

$$J_y = \sigma_{\mathrm{H}} E_x, \qquad J_x = -\sigma_{\mathrm{H}} E_y.$$

Theorem 25.5 (TKNN response · *cited*). *For a clean two-dimensional periodic noninteracting band insulator at zero temperature, with a bulk Fermi-energy gap and isolated occupied bands, the bulk linear Hall response is*

$$\sigma_{\mathrm{H}} = \frac{e^2}{h_{\mathrm{P}}} \sum_{n \ \mathrm{occupied}} C_n.$$

For one filled band the sum has one term.

[1] Their real potential is our $A_{\mathrm{B}} = \mathrm{i}\mathcal{A}$. The extra minus is in their definition $Q = -(1/2\pi) \int \mathrm{d}A_{\mathrm{B}}$, equations (2.38) and (2.41). Their equation (6.5) has the opposite integers, consistent with that continuum definition for the lower band. Their discrete definitions (2.9) and (2.17) instead use the negative sum of forward plaquette arguments divided by 2π, which approaches our C. These source definitions have different signs.

This is the filled-band response theorem of Thouless, Kohmoto, Nightingale and den Nijs (TKNN, 1982). The Kubo-to-curvature step is cited, not proved here. In their p. 406 equation (5), the antisymmetrized derivative integral has the coefficient $ie^2/(2\pi h_{\mathrm{P}})$, matching the curvature normalization above. The response components in this lecture are fixed by the displayed definition.

The sign can also be checked in semiclassical notation. Let the real curvature be $\mathrm{d}A_{\mathrm{B}} = \Omega_{\mathrm{B}}\mathrm{d}k_x \wedge \mathrm{d}k_y$. The wavepacket equations use $\hbar\dot{\mathbf{k}} = -e\mathbf{E}$ and anomalous velocity $-\dot{\mathbf{k}} \times (\Omega_{\mathrm{B}}\mathbf{e}_z)$. Multiplication of this anomalous velocity by charge $-e$ gives a current contribution proportional to $(-E_y, E_x)$. For a filled band the group-velocity integral vanishes by periodicity. The momentum density of states is $\mathrm{d}^2k/(2\pi)^2$, so the anomalous contribution gives

$$\mathbf{J} = \frac{e^2}{\hbar}\int_{\mathbb{T}^2}\frac{\mathrm{d}^2k}{(2\pi)^2}\Omega_{\mathrm{B}}(-E_y, E_x) = \frac{e^2}{h_{\mathrm{P}}}C(-E_y, E_x).$$

This checks the component sign and units of the cited formula. It is not a derivation of the Kubo theorem. For QWZ with only the lower band filled, the four insulating intervals have the conductances 0, $+e^2/h_{\mathrm{P}}$, $-e^2/h_{\mathrm{P}}$ and 0, in increasing u. Indeed, $\min|\mathbf{d}| > 0$ off the critical parameters, so $E_F = 0$ lies in the full gap between $-\min|\mathbf{d}|$ and $+\min|\mathbf{d}|$. At a critical parameter the gap hypothesis fails and the theorem does not assign that quantized value.

25.5 Gap closings and Dirac cones

A *Dirac cone* is a linear two-band crossing with local Hamiltonian

$$H_{\mathrm{D}}(\mathbf{q}) = v_x q_x \sigma_x + v_y q_y \sigma_y + m\sigma_z, \qquad v_x v_y \neq 0.$$

Here $\mathbf{q}$ measures displacement from the crossing momentum. The energies of this local Hamiltonian at $m = 0$ are $\pm\sqrt{v_x^2 q_x^2 + v_y^2 q_y^2}$. The mass m opens a gap. The figure shows the integrated lower-band curvature for unit positive velocities.

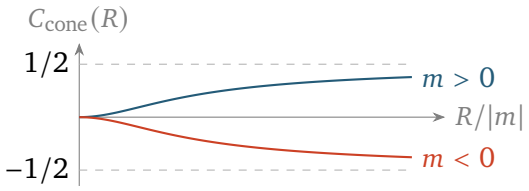


Figure 25.4: The continuum-cone contribution inside a disk of radius R, for $|m| = 1$. The limits are half-integers because the momentum plane is not a closed torus.

Proposition 25.6 (Changes at a closing · *sketched*). *A smooth deformation of an isolated band cannot change its Chern number without losing its isolation. When isolated linear Dirac crossings are the only gap closings, their signed unit changes add. For a cone whose mass increases through zero, the change is* $\operatorname{sgn}(v_x v_y)$.

Sketch. Away from a closing, the eigenline curvature varies smoothly, so the integer remains constant by Proposition 25.2. Near a cone, the degree density gives the plane contribution

$$C_{\mathrm{cone}} = \frac{1}{4\pi}\int_{\mathbb{R}^2}\frac{m v_x v_y \mathrm{d}q_x \mathrm{d}q_y}{(v_x^2 q_x^2 + v_y^2 q_y^2 + m^2)^{3/2}} = \frac{1}{2}\operatorname{sgn}(v_x v_y)\operatorname{sgn}(m).$$

Indeed, rescale the two momentum axes and use polar coordinates. The remaining radial integral is $\int_0^\infty |m| r (r^2 + m^2)^{-3/2} \mathrm{d}r = 1$. Its change across zero is the claimed signed unit. The curvature outside small crossing neighbourhoods has the same limiting integral from either side. In the local rescaling limit each neighbourhood has the displayed unit jump. Disjoint neighbourhoods add their contributions. This local limiting argument sketches the jump. It does not assign an integer to an isolated continuum cone. ∎

For QWZ the velocities at a vertical point are $v_x = \cos k_x$ and $v_y = \cos k_y$. Each local mass increases with u. The jumps are therefore +1 at -2, $-1 - 1 = -2$ at zero and +1 at two, in agreement with the full degree proof.

Self-check. Explain why the QWZ lower-band number is zero at $u = 3$, without counting preimages.

Answer. The component $d_z = 3 + \cos k_x + \cos k_y$ is at least one. Thus the direction map lies in the northern hemisphere and contracts there to the north pole. Its degree is zero by Lecture 14's homotopy invariance.

Summary. An isolated simple Bloch band defines a line bundle on the momentum torus. Its normalized curvature integral is an integer, fixed during any smooth deformation preserving band isolation. For $H = \mathbf{d} \cdot \boldsymbol{\sigma}$ with $\mathbf{d} \neq 0$, the lower-band number is $C = \deg(\widehat{\mathbf{d}})$. For QWZ its values are $0, +1, -1, 0$ in increasing u, with undefined numbers at $-2, 0, 2$. The cited filled-band theorem gives the Hall conductance in units of e^2/h_{P}. The same principle of a homotopy class obstructing deformation will classify defects in Lecture 26.

Reading. Asbóth–Oroszlány–Pályi, *A Short Course on Topological Insulators*, §2.1.3–2.1.4 for the discrete sign; §2.2.1, §2.2.4 and §6.1, pp. 80–82 for QWZ, with continuum $Q = -C$. Thouless et al., *Quantized Hall Conductance in a Two-Dimensional Periodic Potential*, pp. 405–408, especially equation (5) on p. 406.